# Evaluation runs: chan (reliable-channel-aneris)

generated 2026-11-05 from brain-data 0799b9755 2026-11-05 11:44:26 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 882 | conforms | 0.1 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 3 | 48 | 23800 | conforms | 0.7 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 4 | 384 | 539540 | conforms | 7.0 | 4 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 5 | 3840 | 11217600 | conforms | 83.0 | 58 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 6 | 46080 | 221512004 | conforms | 2024.8 | 982 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 7 | 645120 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 704 | 362 | violation | 0.2 | 2 | 1 h, 250 GB | 363 | 2897 | brain03, 2026-11-02 |
| 3 | 48 | 20384 | 15064 | violation | 24.4 | 2 | 1 h, 250 GB | 15065 | 723073 | brain03, 2026-11-02 |
| 4 | 384 | 543060 | 462840 | violation | 1847.5 | 14 | 1 h, 250 GB | 462841 | 177730561 | brain03, 2026-11-02 |
| 5 | 3840 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 5 no-seqno bag, 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag no-seqno, 7 correct bag, 7 bag correct.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 no-seqno bag, 8 no-seqno bag, 8 correct bag, 9 no-seqno bag, 9 correct bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 8 | 882 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 48 | 23800 | conforms | 1.0 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |
| 4 | 384 | 539540 | conforms | 8.6 | 6 | 1 h, 250 GB | brain02, 2026-11-01 |
| 5 | 3840 | 11217600 | conforms | 97.0 | 60 | 48 h, 250 GB | brain05, 2026-11-02 |
| 6 | 46080 | 221512004 | conforms | 2462.0 | 984 | 48 h, 250 GB | brain05, 2026-11-02 |
| 7 | 645120 | 4226823920 | conforms | 53486.0 | 19194 | 48 h, 250 GB | brain01, 2026-11-03 |

### stateful: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 704 | 362 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 48 | 20384 | 15064 | violation | 1.1 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |
| 4 | 384 | 543060 | 462840 | violation | 9.0 | 8 | 1 h, 250 GB | brain02, 2026-11-01 |
| 5 | 3840 | 14061696 | 12864192 | violation | 129.7 | 139 | 48 h, 250 GB | brain05, 2026-11-02 |
| 6 | 46080 | 362542466 | 344704646 | violation | 4067.0 | 2208 | 48 h, 250 GB | brain05, 2026-11-02 |
| 8 | | | | (48 h) | | | 48 h, 250 GB | brain02, 2026-11-03 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): 6 no-seqno bag, 7 bag no-seqno, 7 correct bag, 8 no-seqno bag, 8 correct bag, 9 no-seqno bag, 9 correct bag, 10 no-seqno bag, 10 correct bag, 11 no-seqno bag, 11 correct bag, 12 correct bag, 12 no-seqno bag, 13 correct bag, 13 no-seqno bag.

left out by the table rules (ok): 5 no-seqno bag, 5 correct bag, 6 correct bag.

left out by the table rules (timeout): 6 no-seqno bag, 7 correct bag.

skipped: a smaller size had reached its limit: 7 no-seqno bag, 8 no-seqno bag, 8 correct bag, 9 no-seqno bag, 9 correct bag.

the tool ended with an error (exit-101): 8 bag correct.

left out by the table rules (invalid: foreign load): 9 bag no-seqno, 9 bag correct.

## offline

### offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 882 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 48 | 23800 | conforms | 1.1 | 4 | 10 min, 16 GB | 0.1 | 1.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 384 | 539540 | conforms | 8.7 | 6 | 10 min, 16 GB | 0.1 | 8.7 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 3840 | 11217600 | conforms | 101.7 | 54 | 1 h, 250 GB | 0.4 | 101.7 | 0.0 | merge | brain05, 2026-11-03 |
| 6 | 46080 | 221512004 | conforms | 3464.8 | 978 | 1 h, 250 GB | 1.0 | 3464.6 | 0.1 | merge | brain05, 2026-11-03 |

### offline: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 704 | 6 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 48 | 20384 | 87 | violation | 0.9 | 4 | 10 min, 16 GB | 0.1 | 0.9 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 384 | 543060 | 1092 | violation | 10.8 | 22 | 10 min, 16 GB | 0.1 | 10.8 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 3840 | 14061696 | 14413 | violation | 122.5 | 74 | 1 h, 250 GB | 0.4 | 122.1 | 0.4 | merge | brain05, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 6 no-seqno bag, 7 bag no-seqno, 7 correct bag, 7 bag correct.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 no-seqno bag, 7 correct bag, 8 no-seqno bag, 8 correct bag, 9 no-seqno bag, 9 correct bag.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 882 | conforms | 11.1 | 2031 | 10 min, 250 GB | 0.1 | 11.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 48 | 23800 | conforms | 84.4 | 4566 | 10 min, 250 GB | 0.1 | 84.3 | 0.1 | merge | brain01, 2026-11-03 |

### cross offline, Java implementation against the Rust specification: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 704 | 6 | violation | 6.3 | 205 | 10 min, 250 GB | 0.1 | 6.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 48 | 20384 | 87 | violation | 42.7 | 4312 | 10 min, 250 GB | 0.1 | 42.6 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag no-seqno, 7 bag correct.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 882 | conforms | 4.1 | 171 | 10 min, 250 GB | 4.0 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 48 | 23800 | conforms | 4.4 | 168 | 10 min, 250 GB | 4.3 | 1.1 | 0.1 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 704 | 6 | violation | 4.1 | 171 | 10 min, 250 GB | 4.0 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 48 | 20384 | 87 | violation | 4.4 | 168 | 10 min, 250 GB | 4.3 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag no-seqno, 7 bag correct.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 882 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 48 | 23800 | conforms | 1.3 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 8 | 704 | 362 | violation | 0.1 | 4 | 1 h, 250 GB | 11 | brain01, 2026-11-02 |
| 3 | 48 | 20384 | 15064 | violation | 1.2 | 4 | 1 h, 250 GB | 1 | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag no-seqno, 7 bag correct.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 882 | conforms | 11.2 | 2031 | 10 min, 250 GB | 11.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 23800 | conforms | 84.4 | 4566 | 10 min, 250 GB | 84.3 | 1.1 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 704 | 704 | conforms | 6.2 | 205 | 10 min, 250 GB | 6.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 20384 | 20384 | conforms | 42.6 | 4312 | 10 min, 250 GB | 42.6 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 704 | 6 | violation | 11.1 | 2031 | 10 min, 250 GB | 11.1 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 20384 | 87 | violation | 84.4 | 4566 | 10 min, 250 GB | 84.3 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 882 | conforms | 11.3 | 2031 | 10 min, 250 GB | 0.1 | 11.1 | 0.2 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 23800 | conforms | 84.4 | 4566 | 10 min, 250 GB | 1.1 | 84.3 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 704 | 704 | conforms | 6.2 | 205 | 10 min, 250 GB | 0.1 | 6.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 20384 | 20384 | conforms | 42.8 | 4312 | 10 min, 250 GB | 0.9 | 42.6 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 704 | 6 | violation | 6.2 | 205 | 10 min, 250 GB | 0.1 | 6.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 20384 | 87 | violation | 42.8 | 4312 | 10 min, 250 GB | 1.1 | 42.6 | 0.2 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 882 | conforms | 8.5 | 256 | 10 min, 250 GB | 8.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 23800 | conforms | 52.3 | 5010 | 10 min, 250 GB | 52.0 | 1.1 | 0.3 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 704 | 704 | conforms | 6.0 | 234 | 10 min, 250 GB | 5.9 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 20384 | 20384 | conforms | 41.0 | 4364 | 10 min, 250 GB | 40.9 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 704 | 6 | violation | 8.6 | 256 | 10 min, 250 GB | 8.5 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 20384 | 87 | violation | 52.1 | 5010 | 10 min, 250 GB | 52.0 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 882 | conforms | 8.6 | 256 | 10 min, 250 GB | 0.1 | 8.5 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 23800 | conforms | 52.1 | 5010 | 10 min, 250 GB | 1.1 | 52.0 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 704 | 704 | conforms | 6.0 | 234 | 10 min, 250 GB | 0.1 | 5.9 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 20384 | 20384 | conforms | 40.9 | 4364 | 10 min, 250 GB | 0.9 | 40.9 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 704 | 6 | violation | 5.9 | 234 | 10 min, 250 GB | 0.1 | 5.9 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 23800 | 20384 | 87 | violation | 41.1 | 4364 | 10 min, 250 GB | 1.1 | 40.9 | 0.2 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 5 bag correct, 5 bag no-seqno, 6 bag correct.

planned but never scheduled (grid points beyond the ladder): 6 bag no-seqno.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 882 | conforms | 9.8 | 280 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 23800 | 23800 | conforms | 63.3 | 5055 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 704 | 704 | conforms | 9.1 | 236 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 20384 | 20384 | conforms | 56.8 | 4930 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 704 | 362 | violation | 9.9 | 236 | 1 h, 250 GB | 95 | brain01, 2026-11-03 |
| 3 | 23800 | 20384 | 15064 | violation | 56.5 | 4658 | 1 h, 250 GB | 142 | brain01, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 882 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 23800 | 23800 | conforms | 1.4 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 704 | 704 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 20384 | 20384 | conforms | 1.4 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 882 | 704 | 362 | violation | 0.1 | 4 | 1 h, 250 GB | 10 | brain01, 2026-11-02 |
| 3 | 23800 | 20384 | 15064 | violation | 1.1 | 4 | 1 h, 250 GB | 7 | brain01, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** skipped: a smaller size had reached its limit: 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.
planned but never scheduled (grid points beyond the ladder): 6 bag no-seqno.